THE UNIVERSITY OF HONG KONG
FACULTY OF EDUCATION

Bachelor of Arts and Sciences in Social Data Science [BASc(SDS)]
2024-25

Course Particulars	
Course code:	BSDS3003
Course title:	Data processing and visualization
No. of credits:	6
Course Coordinator	
Name:	Professor LIN Jionghao
Email:	jionghao@hku.hk
Telephone No.:	(852) 39170627
Office:	Room 216, Runme Shaw Building

Course description

This course provides an in-depth exploration of data processing and visualization techniques, teaching students how to develop comprehensive data strategies for generating, collecting, and consuming data. Key topics include data collection and ingestion, data cleaning and pre-processing, data transformation and feature engineering, and data visualization. Through a series of hands-on practices, students will grasp the foundations of building end-to-end data pipelines, and the essential skills needed in exploring, analyzing, and communicating data-driven insights in social data science projects.
Assessment: 100% coursework
Prerequisite: Basic knowledge of Python programming is needed.

Course objectives

Upon completion of the course, students are expected to:

- develop a comprehensive understanding of data processing and visualization techniques.
- develop and implement data strategies for generating, collecting, and consuming data.
- acquire skills to build end-to-end data pipelines and utilize data-driven insights in social data science projects.

Course learning outcomes (CLOs)

Upon successful completion of this course, students should be able to:

	CLOs	**Aligned PLOs**
1	Apply data collection, cleaning, and preprocessing techniques to provide clean datasets ready for analysis	1, 3, 4
2	Construct data pipelines that incorporate data transformation, feature engineering, and integration steps using Python programming	3, 4, 5
3	Analyze data using statistical and computation techniques to draw insights and make informed decisions in social data science contexts	2, 3, 5
4	Develop visualizations that effectively communicate data insights to address societal challenges	1, 4, 6, 7

5	Evaluate the effectiveness of different data strategies in generating, collecting, and transforming data in solving real-world problems	1, 2, 5, 7

Course teaching and learning activities		
Activities	**No. of hours**[1]	**Aligned CLOs**
Lecture	24 (contact hours)	1, 2, 3
Self-study, library search, reading	72	1, 2, 3, 4, 5
Group study	42	1, 2, 3, 4, 5
Online discussion	12	3, 4, 5
Total	150	

Course assessment methods				
Assessment method	**Type of assessment**	**Weighting (%)**	**Aligned CLOs**	**Due Date**
1. Report proposal (Group writing, connecting with final report)	Report proposal 3-4 students in one group, the project aims to use group effort to find public education dataset for data visualization. The proposal should discuss the dataset, potential use of data processing techniques, data analysis methods, and plots.	10%	5	Week 3 (Feb/14/2025)
2. Programming (Individual)	a. Handling missing values, outliers and textual data processing (25%) b. Exploratory data analysis and data visualization (25%)	50%	1, 2, 3, 4, 5	Programming task a (25%): Week 7 March/17/2025 Programming task b (25%): Apr/28/2025
3. Final Report (Group writing, connecting with programming assignments)	Full report (3000 words excluding references with diverse plots) 1. Introduction 2. Dataset description 3. Data processing - If numeric data Missing values Outliers - If textual data Textual data processing 4. Exploratory data analysis 5. Data visualization 6. Potential advanced data visualization	20%	1, 2 3, 4, 5	Week 12 Apr/21/2025
4. Presentation	Presentation and Q&A	20%	3, 4, 5	Week 12

[1] For a 6-credit course, the total no. of teacher-student face-to-face contact (CT) hours should be 24, and the total no. of CT hours and Study Load (SL) hours added together should be in the range of 120 to 150.

(Group presentation, connecting with group writing and programming assignments)				Apr/24/2025
	Total	100%		

Grading criteria

Assessment 1: Programming Tasks

a. Handling missing values, outliers and textual data processing

- **Inspect and Audit Data**: Evaluate the parsed data to identify any existing issues (e.g., missing values and data anomalies)
- **Propose Solutions**: Recommend suitable methods to address and resolve the identified data problems.
- **Data Integration and Normalization**: Integrate data from multiple sources into a unified table and apply normalization techniques to prepare the data for analysis.
- **Iterative Improvement**: Repeat the inspection and auditing process as needed to ensure data quality and completeness.
- **Data Extraction**: Extract data accurately from Excel and PDF formats.
- **Code Quality**: Ensure submitted code runs without errors and produces correct results.
- **Code Structure and Documentation**: Write well-structured, properly commented code that is easy to understand.
- **Method Justification**: Clearly justify the text preprocessing methods used in the Jupyter notebook.
- **Notebook Organization**: Structure the Jupyter notebook logically, demonstrating a clear progression through the assessment tasks.

b. Exploratory data analysis and data visualization

- **Static Visualization Creation**: Demonstrate the ability to create effective static visualizations. using Python.
- **Exploratory Data Analysis**: Perform exploratory data analysis using Python.
- **Identification of Trends and Patterns**: Identify and clearly present trends and patterns observed in the data.
- **Interactive Visualization Implementation**: Implement interactive visualizations using Python.
- **Plot Refinement**: Correct undesired plots into the desired format to improve clarity and presentation.
- **Justification of Plot Choices**: Provide a clear rationale for the refinement
- **Notebook Organization**: Structure the Jupyter notebook logically, demonstrating a clear progression through the assessment tasks.

Assessment 2: Group writing (Report proposal and final report)

Group report (Refer to Assessment Feedback Proforma (Appendix 2))

- Coherent, original, and completed content with relevant and reliable analysis and results
- Demonstrating the knowledge acquired from the course as well as extensive research effort
- Solid academic writing with proper APA referencing format

Please refer to BASc(SDS) Generic Grading Criteria (Appendix 1) and Assessment Feedback Proforma (Appendix 2).

Assessment 3: Presentation and Q&A

Group presentation

- Logic flow and well-organized content
- Clear and aesthetic design of presentation slides
- Being able to capture the interests of audience
- Collaboration: Active, responsible, and respectful communications with group members
- Consistent participation and engagement throughout the project
- Respond to the questions based on the key components from the programming assignments

Course content and topics ***(both topics and schedule are subject to changes)***

Week	Topic	Aligned CLOs
1 Jan/23	Introduction to Data Processing	1, 2, 3, 5

2 Feb/6	Parsing Raw Data	1
3 Feb/13	Lecture with Hands-on Activities: Regular Expression	1
4 Feb/20	Missing Values and Outliers	1, 2
5 Feb/27	Data Integration	1, 2
6 Mar/6	Textual Data Processing	1, 2
7 Mar/20	Introduction to Data Visualization	4
8 Mar/27	Lecture with Hands-on Activities: Python Libraries for Data Visualization	4
9 Apr/3	Exploratory Data Analysis	3
10 Apr/10	Principles of Effective Data Visualization	4, 5
11 Apr/17	Leveraging Large Language Models for Data Processing and Data Visualization	1, 2, 3, 4, 5
12 Apr/24	Group Presentation	1, 2, 3, 4, 5

Recommended readings and online materials *(to be entered in the SIS / Moodle)*

A. Data Processing

[1] McKinney, Wes. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter. Third edition. Sebastopol, CA: O'Reilly Media, Inc., 2022. Print.
[2] Data preparation for machine learning: data cleaning, feature selection, and data transforms in Python
[3] Python Regular Expressions, Python Regular Expressions - part #1

B. Data Visualization

[1] Munzner, T. (2014). Visualization analysis and design. CRC press.
[2] Pajankar, Ashwin. Practical Python Data Visualization: A Fast Track Approach to Learning
[3] Data Visualization with Python. Berkeley, CA: Apress, 2021. Web.
[4] Data Visualization with Tableau Specialization. Coursera, https://www.coursera.org/specializations/data-visualization#courses

Academic Conduct

Plagiarism involves the use of quotations without quotation marks, the use of quotations without indication of the source, the use of another's idea without acknowledging the source, the submission of a paper, report, project, or class assignment (any portion of such) prepared by another person, or incorrect paraphrasing.

Any student who fails due to plagiarism may be referred to the University Disciplinary Committee, which may then recommend discontinuation. Plagiarism is a serious matter. If you have any doubts about whether your use of sources constitutes plagiarism, please refer to this page: https://tl.hku.hk/plagiarism/ or ask your lecturer/course coordinator.

The students are suggested to use the tool Turnitin to check your assignment for correct scholarly practice in citing others' work. You will be provided with a "class ID" and "password" to enable you to check your assignments via appropriate "class" in Turnitin. Further details about Turnitin and how it works are obtainable from http://www.turnitin.com.

The accepted referencing style is APA 7th (http://www.apastyle.org/). The details and examples of APA formatting and style can be found at https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_formatting_and_style_guide/general_format.html.

Student Feedback on Teaching and Learning

At around the end of the semester, the University will invite all students to evaluate courses they take in this semester in terms of course effectiveness and teaching effectiveness. It is an important means for the programs and instructors to improve the course and teaching. The link will be available on Moodle. Please fill it in and give us your comments. Thank you in advance!

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Generic Grading Criteria

Grade & Grade Point	**(+) A (-)** **4.3 4.0 3.7**	**(+) B (-)** **3.3 3.0 2.7**	**(+) C (-)** **2.3 2.0 1.7**	**(+) D** **1.3 1.0**	**F** **0**
Content	• **Excellent** performance • Full coverage of the topic • Meets all basic and higher order goals • Very high level of skills (interpretation, application, analysis, evaluation)	• **Good** performance • Good coverage of the topic • Meets all basic and some higher order goals • High level of skills (interpretation, application, analysis, evaluation)	• **Satisfactory** performance • Reasonable coverage of topic • Meets all basic goals • Satisfactory level of skills (interpretation, application, analysis, evaluation)	• **Pass** • **Borderline acceptable** performance • Limited coverage • Meets only basic goals • Only basic level of skills (interpretation, application, analysis, evaluation)	• **Fail** • **Unsatisfactory** performance • Question / Task possibly misunderstood • Key aspects of topic neglected • Basic goals not met • Basic skills not demonstrated
Logic & Coherence	• Concepts very clearly understood • Argument always logical • Assignment follows highly logical structure and development	• Concepts clearly understood • Argument generally logical • Assignment follows logical structure and development	• Concepts understood • Logic of argument is satisfactory despite some minor weaknesses • Only minor problems with structure or development of assignment	• Basic understanding of concepts • Argument is weak but can be followed • Some problems with structure or development of assignment	• Many concepts not understood • Illogical argument • Assignment rambles & lacks structure
Reading & Research	• Wide, supportive reading of relevant literature • Good critical understanding • Referencing very clear & appropriate	• Well aware of relevant literature • A high level of reflective understanding • Referencing clear & appropriate	• Sound awareness of relevant literature • Evidence of reflective understanding • Referencing satisfactory	• Some awareness of relevant literature • Basic understanding but limited reflection • Some problems with referencing	• Little or no evidence of reading • Uncritical acceptance of others' views • Referencing inadequate
Language	• Expresses meaning very clearly • Highly fluent and accurate use of grammar and vocabulary	• Expresses meaning clearly • Fluent and mostly accurate use of grammar and vocabulary	• Expresses meaning satisfactorily • Satisfactory level of grammar and vocabulary use	• Expression of meaning is acceptable, although: • Some confusion caused by grammatical errors and/or inappropriate vocabulary	• Meaning often unclear • Unacceptable grammar and/or vocabulary use
Presentation	• Very high standard of presentation • Format* requirements fully met	• Good standard of presentation • Format* requirements met	• Satisfactory presentation • Format* requirements mostly met	• Acceptable despite some weaknesses in presentation • Format* requirements partially met	• Poor presentation • Unacceptable format*

**Format includes APA referencing style*

Appendix 2

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Assessment Feedback Proforma

Student Name:	
Course:	
Assessment Task:	

Assessment Feedback *(to be completed by the examiner)*:
(NB The ticks in the various boxes are to provide guidance to students. They are not indicative of weightings towards the final grade. The final grade awarded does not necessarily reflect a simple summation of the ticks as examiner may emphasize one area more than another.)

CONTENT						**CONTENT**
Unsatisfactory performance						**Excellent, outstanding** performance
Question /Task misunderstood						Question /Task very clearly understood
Key aspects of topic neglected						Full coverage of topic
Goals not met (basic or higher order)						All basic and higher order goals met
Basic skills not demonstrated						Very high level of skills
LOGIC & COHERENCE						**LOGIC & COHERENCE**
Concepts not understood						Concepts very clearly understood
Illogical argument						Argument always logical
Assignment rambles & lacks structure						Highly logical structure & development
READING & RESEARCH						**READING & RESEARCH**
Little or no evidence of reading						Wide reading of relevant literature
Uncritical acceptance of other's views						Good critical understanding
Referencing inadequate or inappropriate						Referencing very clear & appropriate
LANGUAGE						**LANGUAGE**
Meaning often unclear						Expresses meaning very clearly
Unacceptable grammar & vocabulary use						Fluent, accurate grammar and vocabulary
PRESENTATION						**PRESENTATION**
Poor presentation						Very high standard of presentation
Unacceptable format						Format requirements fully met

Comments

Examiner: _____ Date: _____